

Enhanced Attention-Based Multi-Layer Perceptron for High-Recall Type 2 Diabetes Prediction Using Clinical Data

Jingying Ng, Jiayu Ong, Junxun Too, and Xinya Siow

Abstract—Type 2 Diabetes Mellitus is a major metabolic disorder that can lead to severe complications if not detected early. This study proposes an enhanced deep learning framework for high-recall diabetes prediction using the Diabetes Prediction Dataset from Kaggle, which contains 100,000 clinical records. Data preprocessing includes One-Hot Encoding, feature standardization, and Synthetic Minority Over-sampling Technique (SMOTE). Three models are evaluated: a baseline Multi-Layer Perceptron, a One-Dimensional Convolutional Neural Network, and a proposed Attention-Enhanced Multi-Layer Perceptron incorporating feature attention and Focal Loss. Preliminary results show that the proposed model achieves the highest recall of 92.0%, outperforming the baseline Multi-Layer Perceptron (86.0%) and 1D-CNN (89.0%). The findings indicate that attention mechanisms and imbalance-aware optimization improve the identification of diabetic individuals and provide an effective framework for early diabetes screening.

Index Terms—Type 2 Diabetes Mellitus, Deep Learning, Multi-Layer Perceptron, Attention Mechanism, Focal Loss, Recall.

I. INTRODUCTION

TYPE 2 Diabetes Mellitus is a chronic metabolic disorder and a major global health issue, often leading to severe complications such as cardiovascular disease, kidney failure, neuropathy, and vision loss. A key challenge is that it progresses silently, with many cases undiagnosed until serious damage occurs. Therefore, early and accurate prediction is critical to reduce long-term morbidity and mortality. Diabetes risk is influenced by clinical, demographic, and lifestyle factors such as HbA1c level, blood glucose, BMI, age, hypertension, heart disease, gender, and smoking history. These variables interact in complex, non-linear ways that traditional models like logistic regression cannot fully capture. Phase 1 exploratory analysis (PCA) also showed overlap between diabetic and non-diabetic samples, indicating the dataset is not linearly separable and requires more expressive models.

Another key issue is severe class imbalance, where diabetic cases make up only about 8.5% of 100,000 records. A naive model may achieve high accuracy by predicting the majority class, but this is misleading in a clinical setting. False negatives are particularly costly because undetected patients may miss early treatment. Therefore, recall (sensitivity) is prioritized over

accuracy, and imbalance-aware methods are essential for a clinically reliable system. This study proposes a Multi-Layer Perceptron (MLP)-based deep learning framework for diabetes prediction using the Kaggle Diabetes Prediction Dataset. The objectives are to review existing deep learning approaches for diabetes prediction, develop an improved imbalance-aware MLP model, and evaluate performance with emphasis on recall for identifying at-risk individuals.

II. LITERATURE REVIEW

A. Review of Existing Deep Learning for Diabetes Prediction

The application of deep learning in diabetes prediction has increased due to its ability to model complex, non-linear relationships among clinical, demographic, and lifestyle factors. Phase 1 exploratory analysis confirmed that the dataset is not linearly separable, motivating the use of neural network-based approaches. However, model selection must be aligned with tabular data characteristics and clinical requirements. Iftikhar et al. proposed EchoceptionNet, an explainable deep learning framework evaluated on the Kaggle Diabetes Prediction Dataset (100,000 records, 8.5% diabetic cases) [4]. The model addresses class imbalance using ProWSyn oversampling and integrates reservoir computing with multi-scale CNN features. It achieved strong performance (accuracy 0.95, recall 0.94, AUC 0.97) with SHAP and LIME for interpretability. However, its architectural complexity and computational cost limit practical implementation compared to simpler models such as MLP. Ayoade et al. compared multiple models, including RNN, DNN, Random Forest, and AdaBoost, across several datasets [2]. Results show no consistent superiority of deep learning models. Performance depends on dataset size and structure, where simpler models may outperform deep learning on smaller datasets due to overfitting. The study also did not address class imbalance or emphasize recall, limiting its clinical applicability. This supports the use of a well-optimized MLP for tabular clinical data. Majyambere et al. proposed a two-stage explainable framework using MLP for diabetes risk prediction [6], focusing on interpretability and clinical decision support. However, imbalance handling and sensitivity optimization were not the primary focus, limiting effectiveness for minority class detection. Farnoosh et al.

introduced DiabetesXpertNet, an attention-based CNN for tabular medical data [3]. The model applies channel attention to prioritize important features such as glucose and insulin. However, the assumption of sequential dependency among clinical variables is debatable, as tabular features are not inherently ordered. Additionally, evaluation on different datasets limits direct comparability.

B. Comparative Analysis of Previous Studies

Across the reviewed studies, a clear progression is evident from standard MLP and DNN architectures toward more complex frameworks incorporating attention mechanisms, explainability modules, and specialized oversampling strategies. However, increasing architectural complexity does not consistently translate into proportional clinical gains, as dataset scale, imbalance handling, and architectural assumptions produce fundamentally different suitability profiles.

The findings of Ayoade et al. directly challenge the assumption that deep learning is inherently superior for clinical prediction. On the smaller PIMA dataset, AdaBoost matched or exceeded RNN and DNN performance, while deep learning showed its advantage only on the larger BRFSS datasets. This reversal reflects a structural mismatch between inductive bias and dataset characteristics: deep learning models require sufficient data volume to learn meaningful representations, and when that volume is absent, additional parameters increase overfitting risk without improving generalization. The 100,000-record Kaggle dataset, while larger than PIMA, remains tabular and low-dimensional at nine features, making architectural simplicity a principled choice rather than a concession.

The suitability argument sharpens when comparing MLP against CNN-based approaches. Convolutional kernels exploit spatial locality among adjacent pixels, but tabular clinical features such as BMI, HbA1c, and blood glucose occupy independent semantic dimensions with no inherent neighbourhood structure. Imposing a spatial inductive bias on such data is actively misleading rather than simply irrelevant. An MLP, treating each feature as an independent input and learning global interaction weights simultaneously, is therefore not simply simpler but arguably more architecturally principled for this data type.

That said, attention mechanisms offer genuine theoretical value. Diabetes risk is not uniformly determined by the same features across all patients: for some profiles, glucose and HbA1c dominate, while for others, BMI and age carry greater discriminative weight. Unlike a standard MLP, attention allows per-instance feature modulation, which is clinically meaningful when patient subpopulations follow distinct risk pathways. However, on a nine-feature dataset, the marginal gain of per-sample weighting diminishes relative to higher-dimensional settings. Class-weighted MLP training addresses the more clinically consequential problem of systematic suppression of the diabetic minority class more directly than attention-based reweighting.

Comparing EchoceptionNet and DiabetesXpertNet reveals two distinct strategies. EchoceptionNet achieves a recall of 0.94 on the same Kaggle dataset through ProWSyn oversampling, making it the most directly comparable benchmark, while

DiabetesXpertNet contributes through attention-driven feature prioritization on different datasets. Critically, EchoceptionNet's high recall is primarily attributable to imbalance correction rather than architectural sophistication, as ProWSyn increases minority class exposure during training whereas attention reweights features without altering the class distribution. Majyambere et al. take a different approach, prioritizing interpretability over sensitivity, which is a valid clinical concern but one that sidesteps the imbalance problem rather than resolving it.

A critical observation across all reviewed studies is that recall is insufficiently prioritized. Ayoade et al. report F1-scores without imbalance handling, Majyambere et al. focus on interpretability over sensitivity, and only EchoceptionNet explicitly optimizes for minority class detection. Since diabetic cases represent only 8.5% of the Kaggle dataset, models without imbalance correction will be biased toward the majority, suppressing sensitivity where it matters most, as a false negative means a diabetic patient is discharged without intervention. Table 1 summarizes the reviewed studies.

C. Research Gap

The reviewed literature reveals three research gaps that this study aims to address. First, recall is insufficiently prioritized as an evaluation metric. In diabetes screening, a false negative carries direct clinical consequences including delayed diagnosis, missed intervention, and accelerated disease progression, yet several reviewed studies report performance primarily through accuracy or F1-score without discussing the recall-precision trade-off or the cost asymmetry between false negatives and false positives.

Second, class imbalance is inconsistently handled. Without deliberate correction, models trained where diabetic cases constitute only 8.5% of instances will be systematically biased toward the majority class. The findings of Ayoade et al. reinforce that deep learning does not consistently outperform simpler approaches on tabular clinical data unless imbalance is explicitly accounted for, meaning reported accuracy figures in several reviewed studies overstate true clinical reliability.

Third, a gap exists between model complexity and architectural justification for tabular clinical data. Attention mechanisms offer genuine theoretical value through per-sample feature prioritization, and CNN-based architectures achieve strong results, but these gains carry substantial implementation overhead and rest on spatial or sequential inductive biases absent from tabular biomarker data. A well-optimized MLP with class-weighted loss correction is more architecturally principled and computationally accessible for structured low-dimensional datasets. Although EchoceptionNet addresses imbalance and DiabetesXpertNet addresses feature prioritization, no reviewed study simultaneously prioritizes recall as the primary metric, applies loss-based imbalance correction rather than synthetic oversampling, and maintains full architectural alignment with the independent-feature structure of tabular clinical data. It is this combination that the present study seeks to provide.

TABLE 1
SUMMARY OF PREVIOUS STUDIES

Source	Dataset Size	Model Type	Performance	Strengths	Limitations
Ifitikhar et al. (2025)	Kaggle Diabetes Prediction Dataset (100,000 records)	EchoceptionNet (Hybrid temporal-spatial deep learning)	Accuracy: 0.95, Recall: 0.94, F1: 0.95, AUC: 0.97	Uses the same dataset for direct comparison Handles class imbalance with ProWSyn Integrates SHAP and LIME for explainability	Complex dual-module architecture
Ayoade et al. (2025)	PIMA, BRFSS Survey, BRFSS Subset, EMR datasets	MLP, DNN, RNN, ensemble models (AdaBoost, Random Forest)	Deep learning outperformed on larger datasets, machine learning matched deep learning on smaller datasets	Demonstrates that model performance depends on dataset scale and inductive bias fit Cross-dataset perspective informs architecture selection	No imbalance correction applied Recall not optimized Limits clinical reliability for minority class detection
Majjamber et al. (2026)	Large clinical dataset	Two-stage explainable machine learning including MLP	High interpretability and strong predictive performance	Emphasizes clinical transparency and risk stratification	Focuses on interpretability rather than sensitivity or imbalance handling
Farnoosh et al. (2025)	PIMA Indian Diabetes Dataset + Frankfurt Hospital Germany Diabetes Dataset	DiabetesXpertNet (Attention-based CNN for tabular data)	High predictive performance across both datasets	Uses attention mechanisms to prioritize important biomarkers	Evaluated on different datasets from this study Sequential assumption may be unsuitable for tabular features

III. METHODOLOGY

A. Selected Models for Comparison

Based on the diabetes prediction literature reviewed and the dataset characteristics, two models were selected for comparative evaluation in this study: the Multi-Layer Perceptron (MLP) and the One-Dimensional Convolutional Neural Network (1D-CNN).

The MLP serves as the primary baseline model, consisting of fully connected hidden layers with non-linear activation functions and a sigmoid output layer. The fully connected architecture is particularly suited for tabular clinical datasets, where features such as HbA1c level, BMI, and blood glucose level represent independent biological measurements with no inherent sequential relationship [2]. Unlike spatial or time-series data, tabular features carry no neighbourhood structure, making global feature interaction learning through fully connected layers a more principled choice than locality-based operations [2].

The selection of MLP is further reinforced by Ayoade et al., who demonstrated that model performance is tightly coupled to dataset scale and feature structure, and that a well-optimised MLP remains a principled architecture choice for tabular clinical datasets where features represent independent biological measurements without inherent sequential relationships [2]. The Principal Component Analysis conducted further confirmed that the dataset is not linearly separable, justifying the use of a non-linear neural network model. From a computational perspective, the MLP is efficient for datasets with a relatively small number of features, as its parameter

count scales with the number of input features and hidden units, making it practical for moderate-scale academic implementation [5].

The 1D-CNN is selected as the comparative deep learning model to provide a meaningful architectural contrast against the MLP. Unlike the MLP, which processes all input features simultaneously through fully connected layers, the 1D-CNN treats the 12 preprocessed tabular features as a one-dimensional sequence and applies convolutional filters across this sequence to detect local feature interaction patterns. The selection of 1D-CNN is supported by Farnoosh et al., who demonstrated that attention-based CNN architectures can achieve strong predictive performance on structured clinical tabular datasets, though direct comparison is not possible as DiabetesXpertNet was evaluated on different datasets [3].

Several limitations of applying 1D-CNN to tabular clinical data should also be acknowledged. The convolutional mechanism assumes that features positioned adjacent to one another in the input sequence share stronger local relationships than non-adjacent features [2]. In clinical tabular data, this assumption may not be biologically meaningful, as relationships between variables such as HbA1c level and blood glucose level exist regardless of their position in the feature vector. Furthermore, feature ordering in the preprocessed dataset is determined by the encoding process rather than a clinically meaningful sequence, which limits the biological interpretability of learned convolutional patterns. From a computational perspective, the 1D-CNN introduces additional parameters through its convolutional and pooling layers, resulting in higher overhead compared to a comparably sized MLP for a 12-feature dataset [3]. Together, the MLP and 1D-CNN provide a structured basis for comparing two

fundamentally different deep learning approaches on the same preprocessed dataset, with both models subsequently evaluated against the proposed enhanced model.

B. Enhanced Model Proposal

The proposed Attention-Enhanced Multi-Layer Perceptron (Attention-MLP) extends the baseline MLP through a feature-level attention mechanism alongside targeted regularization and optimization strategies, motivated by the three research gaps identified

In diabetes screening, recall is a more clinically critical metric than overall accuracy. A model predicting all cases as non-diabetic achieves 91.5% accuracy on this dataset yet identifies no diabetic patient. False negatives carry severe clinical consequences, including delayed diagnosis and irreversible complications such as diabetic neuropathy, nephropathy, and cardiovascular disease [1], [10]. The clinical cost of a false negative far outweighs that of a false positive, which results only in a confirmatory follow-up [4]. Therefore, the Attention-MLP is optimized primarily for high recall, accepting a moderate precision trade-off as clinically justified [5].

To address the 8.5% class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training set to generate synthetic diabetic samples through interpolation between existing minority class instances. SMOTE was selected over random oversampling, which duplicates samples without adding diversity, random undersampling, which discards valuable majority class data, and cost-sensitive learning, which offers less explicit control over minority class distribution [8]. However, SMOTE's linear interpolation may not fully represent true clinical diversity of diabetic patients, introducing a risk that the model learns patterns specific to synthetic interpolated samples rather than genuine clinical distributions, potentially reducing generalization to real-world patient data [8]. To mitigate this overfitting risk, Dropout regularization is therefore incorporated to prevent overfitting on synthetic samples by randomly deactivating neurons during training, encouraging generalized representations that transfer to real distributions.

A soft attention layer is introduced immediately following the input layer. This mechanism learns a set of continuous attention weights over the 12 input features, dynamically emphasizing clinically critical biomarkers such as HbA1c level and blood glucose level while suppressing less discriminative features, informed by Farnoosh et al. [3]. While the marginal benefit of attention is conditional on dataset dimensionality [3], the heterogeneous nature of the 100,000-record dataset, where patient subpopulations may follow distinct risk pathways, provides sufficient motivation for per-sample feature prioritization.

To further address class imbalance at the optimization level, the standard Binary Cross-Entropy loss is replaced with Focal Loss, which dynamically downweights the contribution of easily classified non-diabetic samples and concentrates the training signal on the harder-to-identify diabetic minority class [9]. This complements the SMOTE technique by addressing

imbalance simultaneously at both the data level and the loss function level, providing a more robust combined strategy for minority class detection. Batch Normalization is applied after each fully connected hidden layer to normalize activations, reduce internal covariate shift, and accelerate convergence. A ReduceLROnPlateau learning rate scheduler is employed to automatically reduce the learning rate when validation loss plateaus, complemented by early stopping to terminate training when validation performance no longer improves. The complete architecture of the Attention-MLP is summarized in Table 2.

TABLE 2
ARCHITECTURE AND ENHANCEMENT COMPONENTS OF THE ATTENTION-MLP

Component	Description
Input Layer	Accepts 12 standardized float32 features
Feature Attention Layer	Learns soft weights over input features to emphasize critical biomarkers
Hidden Layers	Fully connected layers with ReLU activation
Batch Normalization	Normalizes activations after each hidden layer to stabilize training
Dropout	Reduces overfitting on SMOTE-balanced training data
Output Layer	Single sigmoid neuron for binary diabetes classification
Loss Function	Focal Loss to address class imbalance at the optimization level
LR Scheduler	Reduce LROnPlateau for adaptive learning rate adjustment
Early Stopping	Halts training when validation performance plateaus

The Attention-MLP is expected to improve recall by combining SMOTE, Focal Loss, and Dropout, provide clinically interpretable biomarker weighting through attention scores and demonstrate that meaningful performance gains are achievable without the computational overhead of complex frameworks such as DiabetesXpertNet or EchoceptionNet, directly addressing the complexity trade-off identified across the reviewed studies.

IV. FRAMEWORK DESIGN

A. Framework Overview

The proposed framework is an end-to-end deep learning pipeline for diabetes prediction using clinical, demographic, and lifestyle variables from the Diabetes Prediction Dataset. The data is cleaned, encoded, scaled, and split into training, validation, and testing sets. SMOTE is applied only to the training set to balance diabetic and non-diabetic samples, while the validation and testing sets remain untouched and imbalanced to prevent data leakage and reflect realistic clinical conditions. Three models are trained and compared: baseline MLP, 1D-CNN, and Attention-MLP. The Attention-MLP includes feature attention, Batch Normalization, Dropout, Focal Loss, adaptive learning rate scheduling, and Early Stopping. The framework also includes attention-weight extraction, threshold optimization, and K-Fold Cross-Validation to improve interpretability, select a suitable screening threshold, and evaluate model robustness.

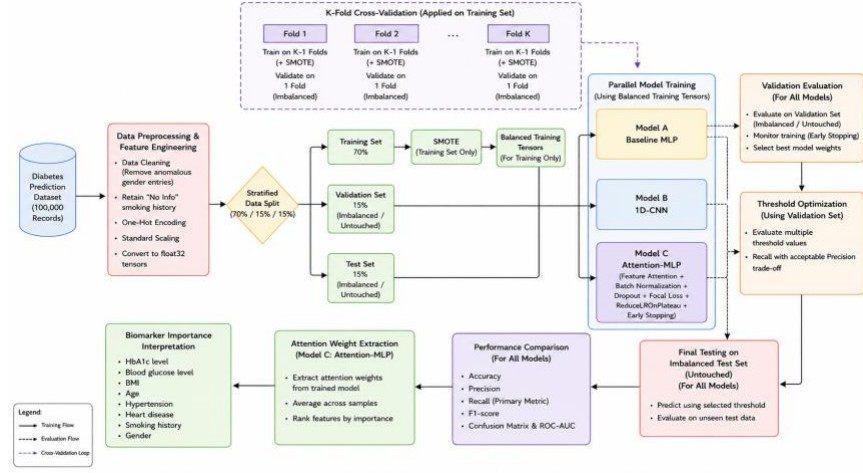


Fig. 1. End-to-End Deep Learning Framework for Diabetes Prediction

B. Framework Diagram

Figure 1 shows the end-to-end deep learning framework of prediction model.

C. Workflow Explanation

Step 1: Input Dataset

The workflow begins with the Diabetes Prediction Dataset, which contains 100,000 records with demographic, lifestyle, and clinical features such as age, gender, BMI, smoking history, hypertension, heart disease, HbA1c level, and blood glucose level. The target variable is diabetes status.

Step 2: Data Preprocessing / Feature Engineering

The dataset is cleaned, encoded using One-Hot Encoding, scaled using Standard Scaling, and converted into float32 tensor format. It is then split into training, validation, and testing sets using stratified partitioning. SMOTE is applied only to the training set, while the validation and testing sets remain untouched and imbalanced for realistic evaluation.

Step 3: Model A Pipeline – MLP

The baseline MLP receives all twelve preprocessed features and uses fully connected layers with ReLU activation to learn non-linear relationships among clinical and lifestyle variables. A sigmoid output layer is then used to produce the binary diabetes prediction.

Step 4: Model B Pipeline – 1D-CNN

The 1D-CNN treats the tabular features as a one-dimensional sequence and applies convolutional filters to capture local feature interactions before passing the extracted patterns into fully connected layers for final classification. This model is used to compare whether convolutional feature extraction improves prediction performance.

Step 5: Model C Pipeline - Attention-MLP

The Attention-MLP uses feature attention, Focal Loss, and regularization to improve diabetes prediction. The attention layer assigns weights to input features so the model can focus more on important clinical variables.

Step 6: Threshold Optimization

Since the default threshold may not be ideal for diabetes screening, threshold optimization is used to choose a suitable classification threshold based on validation performance. The selected threshold prioritizes high Recall while maintaining an acceptable number of false positives.

Step 7: Attention Weight Extraction

After training, attention weights are extracted from the Attention-MLP. These weights are averaged across samples to identify the most influential features in diabetes prediction. This provides biomarker-level interpretation instead of only reporting prediction accuracy.

Step 8: K-Fold Cross-Validation

To test model robustness, the training data is divided into several folds, where each fold is used once for validation while the remaining folds are used for training. The final results are reported using the average and standard deviation of Accuracy, Precision, Recall, and F1-score.

Step 9: Final Performance Evaluation

All models are evaluated using the untouched imbalanced testing set. Recall is the primary evaluation metric because it measures the model's ability to correctly identify diabetic patients. Precision, F1-score, Accuracy, and confusion matrix results are also reported to provide a comprehensive performance assessment.

V. FINAL RESULTS AND DISCUSSION

A. Comparative Predictive Performance

The three deep learning pipelines were evaluated on a completely isolated testing set to reflect genuine clinical distributions. The baseline Multi-Layer Perceptron (MLP) achieved the highest overall accuracy across the entire feature space at 93.0%. However, it produced the weakest clinical screening performance yielding a minority class Recall of only 85.0%.

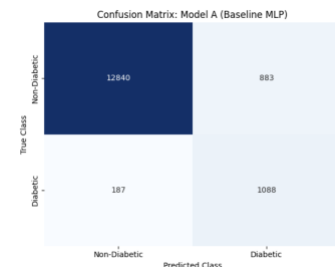


Fig. 2. Confusion Matrix of Baseline MLP

The One-Dimensional Convolutional Neural Network (1D-CNN) utilized localized sequence filters to capture structural interactions and raising the minority Recall to 88.0%.

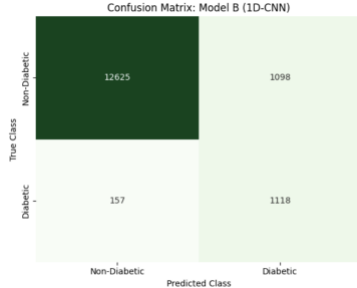


Fig. 3. Confusion Matrix of 1D-CNN

The proposed Enhanced Attention-MLP clearly outperformed both baselines by achieving a peak minority Recall of 92.0%. Although the strict optimization on the minority class caused a corresponding drop in Precision (46.0%) and overall Accuracy (90.0%), this outcome represents an acceptable trade-off in medical diagnostic screening. The clinical and long-term financial costs of generating a false negative by allowing type 2 diabetes to progress unchecked outweigh the cost of a false positive, which simply triggers a secondary diagnostic check. By minimizing false negatives the Attention-MLP confirms its effectiveness as a reliable frontline screening mechanism.

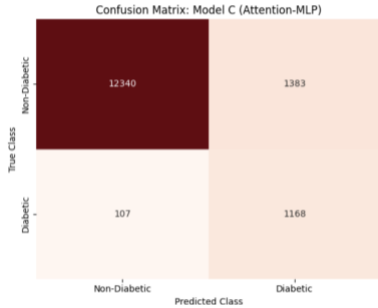


Fig. 4. Confusion Matrix of Attention-MLP

B. Robustness Validation

To validate that the Attention-MLP's high Recall was not an artifact of dataset partitioning or synthetic data leakage, a 5-Fold Cross-Validation was conducted. Crucially, SMOTE was applied strictly within the training folds during each iteration to preserve the integrity of the validation sets.

The Attention-MLP demonstrated exceptional stability, achieving a Mean Minority Recall of $90.77\% \pm 0.88\%$ across all folds. Furthermore, the model proved highly computationally efficient. The average training time across the entire pipeline was just 74.63 seconds per fold, with a peak memory footprint of 704.33 MB. This confirms that significant predictive gains can be achieved through targeted architecture enhancements without requiring the massive computational overhead associated with complex frameworks like DiabetesXpertNet.

C. Clinical Interpretability via Attention Mechanisms

A major limitation of traditional neural networks in healthcare is their "black box" nature. To address this, the soft-attention layer weights were extracted during inference to observe how the network dynamically prioritizes clinical variables.

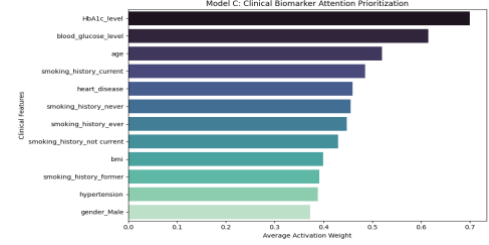


Fig. 5. Clinical Biomarker Attention Prioritization extracted

The attention weight distribution confirms that the model aligns with established medical literature. The network automatically assigned the highest continuous activation weights to critical biomarkers such as HbA1c levels and blood glucose levels dynamically scaling down the impact of less discriminative demographic features. Unlike static feature selection, this mechanism retains all biological information while providing clinicians with transparent reasoning behind the model's individual risk assessments.

D. Threshold Optimization and Precision-Recall Trade-off

Standard binary classification models default to a 0.50 decision threshold. However, clinical systems require threshold tuning to match the specific risk tolerance of the screening environment. A Precision-Recall (PR) Curve was generated to evaluate the Attention-MLP's operational flexibility.

At the default threshold of 0.50, the model heavily prioritizes Sensitivity (Precision: 0.458, Recall: 0.916), ensuring maximum identification of at-risk patients. Mathematical threshold optimization (maximizing the F1-Score) identified an optimal diagnostic threshold of 0.6672. Shifting the decision boundary to this threshold dramatically increases Precision to 0.956 while reducing Recall to 0.687. This demonstrates that the proposed framework is highly adaptable. Healthcare administrators can dynamically slide the threshold to prioritize maximum catch rates (Recall) for initial population screening, or prioritize high confidence (Precision) for environments with limited secondary testing resources.

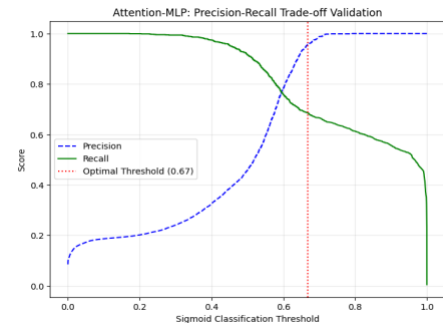


Fig. 6. Precision-Recall Trade-off Validation for Attention-MLP.

E. Conclusion

This study successfully engineered and evaluated an enhanced deep learning framework for type 2 diabetes screening by bridging the gap between high predictive performance and practical clinical deployment. By evaluating a baseline Multi-Layer Perceptron (MLP), One-Dimensional Convolutional Neural Network (1D-CNN) and the proposed Attention-Enhanced MLP on a dataset of 100,000 clinical records, this research addressed critical limitations in current medical AI applications.

The findings demonstrate that traditional accuracy metrics often mask severe deficiencies in imbalanced medical datasets. By integrating Synthetic Minority Over-sampling Technique (SMOTE) with Focal Loss optimization, the proposed Attention-MLP successfully prioritized the minority diabetic class and achieving a peak Recall of 92.0%. This significantly minimizing the risk of undetected false negatives. Furthermore, the 5-Fold Cross-Validation confirmed the model's robustness and prove it does not overfit to synthetic data.

Crucially, this framework overcomes the "black box" limitation of traditional neural networks. The extraction of dynamic soft-attention weights provided transparent clinical interpretability. This verify that the model autonomously prioritizes critical biological markers such as HbA1c and blood glucose levels. Coupled with mathematical threshold optimization to balance the Precision-Recall trade-off, the Attention-MLP offers healthcare administrators a highly adaptable and resource-efficient screening tool that can be dynamically tuned to match the specific risk tolerances of diverse medical environments.

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REFERENCES

- [1] American Diabetes Association. (2023). Classification and diagnosis of diabetes: Standards of medical care in diabetes—2023. *Diabetes Care*, 46 (Supplement 1), S19-S40. <https://doi.org/10.2337/dc23-S002>
- [2] Ayoade, O. B., Shahrestani, S., & Ruan, C. (2025). Machine learning and deep learning approaches for predicting diabetes progression: A comparative analysis. *Electronics*, 14(13), 2583. <https://doi.org/10.3390/electronics14132583>
- [3] Farnoosh, R., Abnoosian, K., Isewid, R. A., & Javaheri, D. (2025). DiabetesXpertNet: An innovative attention-based CNN for accurate type 2 diabetes prediction. *PloS one*, 20(9), e0330454. <https://doi.org/10.1371/journal.pone.0330454>
- [4] Iftikhar, K., Javaid, N., Ahmed, I., & Alrajeh, N. (2025). A novel explainable deep learning framework for accurate diabetes mellitus prediction. *Applied Sciences*, 15(16), 9162. <https://doi.org/10.3390/app15169162>
- [5] Kavakiotis, I., Tsave, O., Salifoglou, A., Maglaveras, N., Vlahavas, I., & Chouvarda, I. (2017). Machine learning and data mining methods in diabetes research. *Computational and Structural Biotechnology Journal*, 15, 104-116. <https://doi.org/10.1016/j.csbj.2016.12.005>
- [6] Majyambere, S., Lindgren, T., Twizere, C., & Ntakirutimana, I. (2026). Early Type 2 diabetes risk prediction using explainable machine learning in a two-stage approach. *Frontiers in Digital Health*, 8, 1743619. <https://doi.org/10.3389/fdgth.2026.1743619>
- [7] Mustafa, T. (2023). *Diabetes prediction dataset* [Data set]. Kaggle. <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>
- [8] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002. <https://doi.org/10.1613/jair.953>
- [9] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2980–2988. <https://doi.org/10.1109/ICCV.2017.324>
- [10] Zheng, Y., Ley, S. H., & Hu, F. B. (2018). Global aetiology and epidemiology of type 2 diabetes mellitus and its complications. *Nature Reviews Endocrinology*, 14(2), 88-98. <https://doi.org/10.1038/nrendo.2017.151>